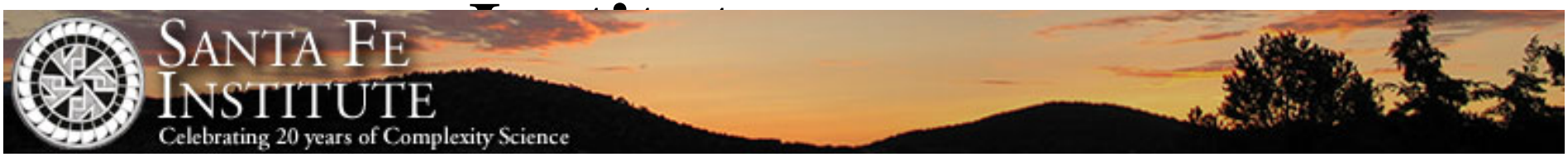
sihT esraP oT yrT

Two Quantum Tapas



Dave Bacon

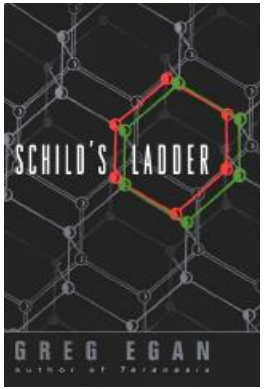
Santa Fe





Henry Everitt (ARO, paraphrase)

"Cumulative worldwide quantum information science spending is now over 1 Billion dollars."



Greg Egan (Schild's Ladder)

"Cass dated the advent of civilization to the invention of the quantum singleton processor. The Qusp."

Tapas Menu

Valued-Added Multipartite Quantum Correlations……..**>**12.5 ¢

(Trace Tessier, Carl Caves, Ivan Deutsch, Bryan Eastin, db)

A Quantum Error Correcting Subsystem….2 ¢

(db
)

Dihedral Hidden Subgroup Problem……query for $

(Andrew Childs, Wim van Dam, db)

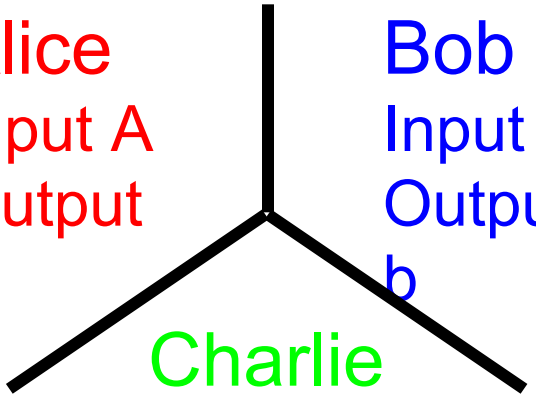
Efficient Quantum Clebsch-Gordan Transforms……..$1 per n

per d

per log(ε-1)

(Isaac Chuang, Aram Harrow, db)

GHZ/Mermin Game



Input: X or Y, Output: +1 or -1

Win condition:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

Classical

aY bY cX = -1

aY bX cY = -1

aX bY cY = -1

× ——————

aX bX cX = -1

➡ not possible to always win

Quantum

Quantum FIREBALL

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

XXX|GHZ⟩ =|GHZ⟩

YYX|GHZ⟩ =|GHZ⟩

➡ possible to always win

Cheaters Always Win



Alice
Input A
Output
a

Bob
Input B
Output
b



Charlie
Input C
Output c

Win condition:

A	B	C	abc
X	X	X	+1
Y	Y	X	-1
Y	X	Y	-1
X	Y	Y	-1

Charlie: always outputs c=+1

Bob: always outputs b=+1, tells Alice his measurement

Alice: knows A and B

A	B	a	abc
X	X	+1	+1
Y	Y	-1	-1
Y	X	-1	-1
X	Y	-1	-1

Using a single bit of communication they can win.

GHZ Extended

Alice Input A Output a	Bob Input B Output b
Charlie Input C Output c	Dodo Input D Output d

win condition

A	B	C	D	abcd
X	X	X	X	+1
X	X	Y	Y	-1
X	Y	Y	X	-1
X	Y	X	Y	-1

Quantum

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$$

possible to always win

Classical (no talking)

aX bX cY dY= -1

aX bY cX dY= -1

aX bY cY dX= -1

× ——————

aX bX cX dX= -1

impossible to always win

Classical (A and B talk)

vXXcY dY= -1

vXYcY dX= -1

vXYcX dY= -1

× ——————

vXXcX dX= -1

impossible to always win

GHZ Super Extended

win condition

A	B	C	D	abcd
X	X	X	X	+1
X	X	Y	Y	-1
X	Y	Y	X	-1
X	Y	X	Y	-1
Y	X	X	Y	-1
Y	X	Y	X	-1
Y	Y	X	X	-1

Quantum

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle)$$

possible to always win

Alice
Input A
Output a

Bob
Input B
Output b

Charlie
Input C
Output c

Dodo
Input D
Output d

If any two parties talk, they cannot always win!

Bell Inequality With Communication

two parties, three two-outcome measurements
local hidden variable theory + 1 bit of communication

D. Bacon and B.F. Toner, Bell Inequalities with Auxiliary Communication
PRL **90**, 157904 (2003)

$$|-ab+ac-ba+bb+bc+ca+cb+cc| \le 6$$

$$|aa+2ab-2ac+2ba+bb+2bc-2ca+2cb+cc| \le 11$$



but, quantum theory has

$$\langle -AB+AC-BA+BB+BC+CA+CB+CC \rangle \quad QM \le 6$$

$$\langle AA+2AB-2AC+2BA+BB+2BC-2CA+2CB+CC \rangle$$

$$QM \le 11$$

four parties, two two-outcome measurements
local hidden variable theory + 1 bit of communication

$$|aaaa-aabb-abab-abba-baab-baba-bbaa| \le 5$$



quantum theory violates this bound

$$\langle AAAA-AABB-ABAB-ABBA-BAAB-BABA-BBAA \rangle$$

Measure It!

Just as Bell Inequalities tell us that we can experimentally distinguish between quantum theory and local hidden variable models,

this Bell Inequality with Communication tells us that we can experimentally distinguish between quantum theory and local hidden variable models, plus one bit of cheating communication.

$$|aaaa-aabb-abab-abba-baab-baba-bbaa| \leq 5$$

$$\langle AAAA-AABB-ABAB-ABBA-BAAB-BABA-BBAA \rangle_{GHZ} = 7$$

3,4,…,n

win condition

A …….	n-3	n-2	n-1	n	abcd
X …….	X	X	X	X	+1
X …….	X	X	Y	Y	-1
X …….	X	Y	Y	X	-1
X …….	X	Y	X	Y	-1

local hidden variable theory
plus n-3 bits of communication

$|A\ldots A - \sum(\text{observables with 2 B's and rest A's})| \leq n(n-1)/2 - 1$

$\langle A\ldots A - \sum(\text{observables with 2 B's and rest A's})\rangle$ GHZ $= n(n-1)/2+1$

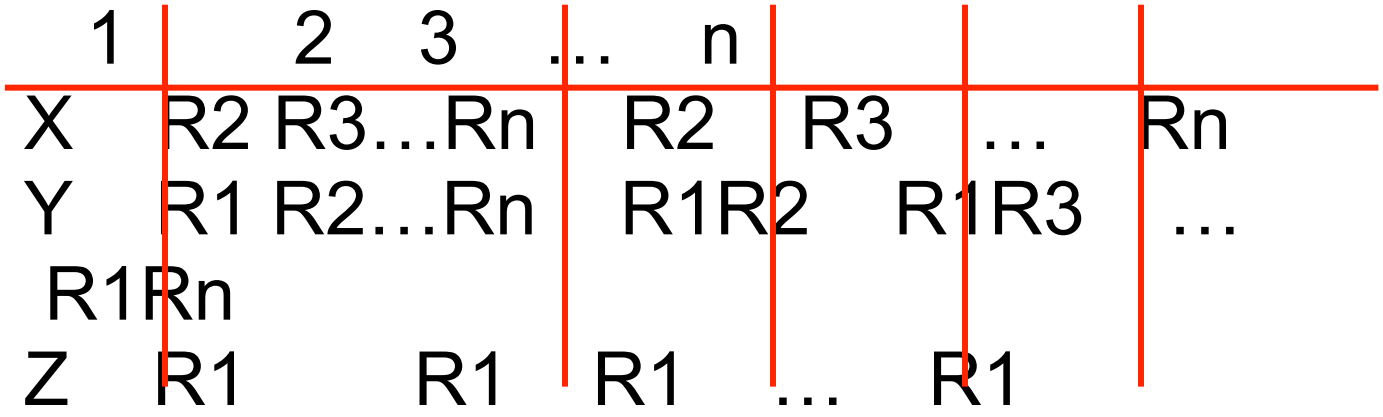
Bell Inequality with n-3 bits of communication

Simulating Pauli Measurements on GHZ

n parties, each measure {X,Y,Z} on n party GHZ state

BIwC implies least n-2 bits of communication necessary

n-2 bits of communication is also sufficient



$R_i \in \{+1,-1\}$

uniform random variable s

parties 2…n-1 communicate to party 1 whether they measured Y or not

party 1 flips her outcome bit if [hers Y plus 2…n-1 Y's] is odd else she does nothing

GHZ and Determinism

Peres: ““The list of authors [claiming that a single experiment is sufficient to invalidate local realism] is too long to be given explicitly, and it would be unfair to give only a partial list.”

$$|aa+ab+ba-bb|\leq 2$$

CHSH inequality

each party outputs uniform random outcome irrespective of what they are measuring

measure each observable once (four experiments)

with probability = 1/8, $|aa+ab+ba-bb|=4$

similar argument for GHZ paradox, with probability 1/8,

$$|-aaa+abb+bab+bba|=4$$

GHZ and Kochen-Specker

Einstein, Podolsky, Rosen element of reality:

"If, without in any way disturbing a system, we can predict with certainty (i.e., with probability equal to unity) the value of a physical quantity, then there exists an element of physical reality corresponding to this physical quantity.

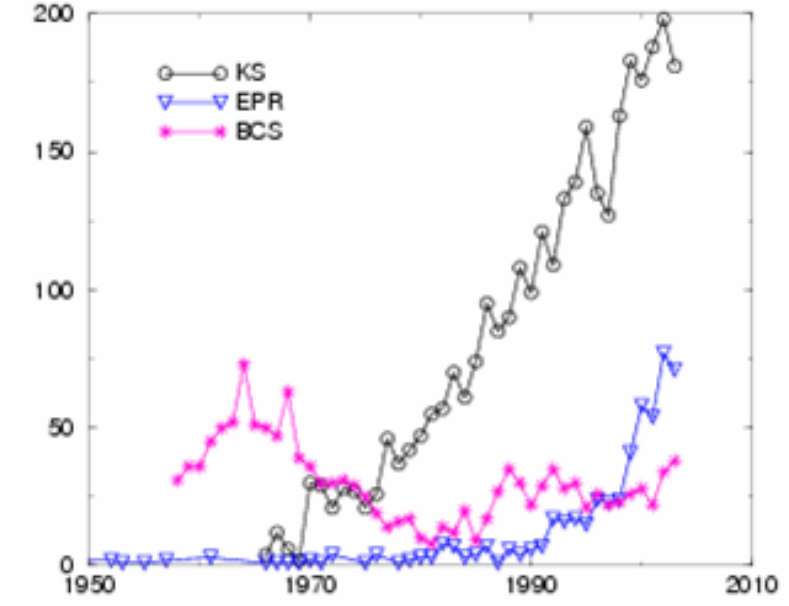


FIG. 9: Citation history of 3 classic highly-cited publications. Each is identified by author initials (see text).

quantum computing saved EPR!

from S. Redner physics/040713

Cits



cits not mits

Kochen-Specker-GHZ-Mermin contradiction:

v(X1)v(X2)v(X3)v(X1X2X3)=1
v(Y1)v(Y2)v(X3)v(Y1Y2X3)=1
v(Y1)v(X2)v(Y3)v(Y1X2Y3)=1
v(X1)v(Y2)v(Y3)v(X1Y2Y3)=1
v(X1X2X3)v(Y1Y2X3)v(Y1X2Y3)v(X1Y2Y3)=-1

GHZ elements of (Pauli) reality:

1	2	3	…	n			
X	R2	R3…Rn		R2	R3	…	Rn
Y	R1	R2…Rn		R1R2	R1R3		…
R1Rn							
Z	R1	R1		R1	…	R1	

+ n-2 bits of communication

cits (contextual bits): number of bits communicated from system to measurement device to restore non-contextuality.

The Finiteness Conjecture

Conjecture: all multipartite entangled systems can be exactly simulated by augmenting a local hidden variable theory with an amount of communication polynomial in the product of the dimensions of the entangled subsystems.

- projective measurements on two entangled qubits (Brassard, Cleve, Tapp 99, Toner, Bacon 03)
- all bipartite two outcome correlation observables (Bacon, Toner 05)
- Knill-Gottesman States and Pauli Measurements (Gottesman, Knill 97)
- optimal solution for Pauli measurements on GHZ (Tessier, Caves, Deutsch, Eastin, Bacon 04)



Multipartite Sausage Correlations

Subspaces Versus Subsystems

The most general way to encode quantum information is into subsystem (not subspaces)



two

c $\mathcal{H}_2 \otimes \mathcal{H}_2$

<u>encode a qubit in a subspace</u>

$|0_L\rangle = |1\rangle \otimes |1\rangle$

$|1_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle - |1\rangle \otimes |0\rangle)$

$\alpha|0_L\rangle + \beta|1_L\rangle$

<u>encode a qubit in a subsystem</u>

$(\alpha|0_L\rangle + \beta|1_L\rangle) \otimes |\psi\rangle$

generic subsystem:

$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$

Quantum Error Correcting Subsystems

Knill and Laflamme, “A theory of quantum error correcting codes” PRA, 55, pp. 900-911 (1997)

Knill, Laflamme, and Viola, “Theory of quantum error correction for general noise,” PRL, 84, pp. 2525-2528 (200)

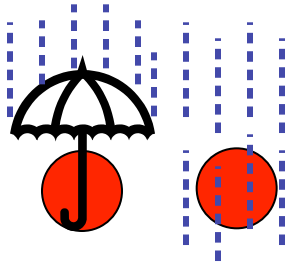
Subsystem encoding:

•implicit from the beginnings of quantum error

•explicit for degenerate (noiseless) codes and dynamic recoupling

Quantum error correcting subsystem codes?

Subsystem Error Correction



Only need to protect subsystem,
not full subspaces.

$$\mathcal{H}_{sys} = (\mathcal{C} \otimes \mathcal{D}) \oplus \mathcal{P}$$

encoding subsystem

$$(|\phi\rangle\langle\phi| \otimes \rho_D) \oplus 0 \quad \mathcal{E} \quad \mathcal{R} \quad (|\phi\rangle\langle\phi| \otimes \tilde{\rho}_D) \oplus 0$$

encoded quantum information

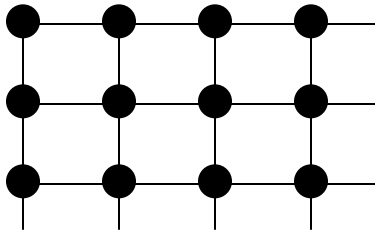
error

recovery

recovery need not perfectly restore system, but only preserve information encoded into subsystem.

Flatland

n by n cubic lattice of qubits:



$$X_{2,2} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right] \quad \begin{matrix} & & & & X & & & \\ & & & & & & & \end{matrix} \quad \begin{matrix} & & & & Z & & & \\ & & & & & & & \end{matrix}$$

$$Z_{2,1} = \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

$$Y_{2,1} = i \left[\begin{array}{cccc|cccc} 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ \vdots & & & \vdots & \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{array}\right]$$

Generic Pauli group operator on cubic lattice:

$$P = i^k \left[\begin{array}{c|c} A & B \end{array}\right]$$ A, B n by n 0,1 matrices

X Columns, Z Rows

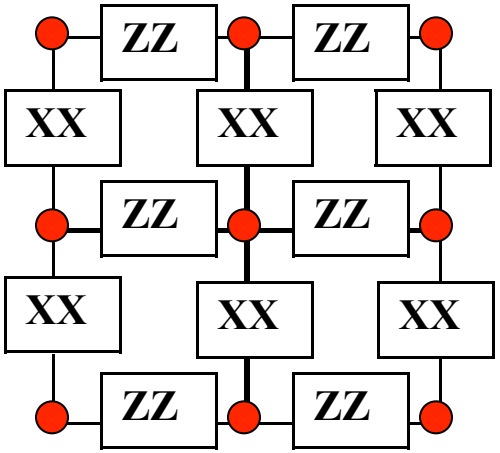
X Z

$$\left[\begin{array}{c|c} A & B \end{array}\right]$$

$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

$$c\left[\begin{array}{cccc|cccc} 0&1&0&0&1&1&1&1 \\ 0&1&0&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&1&1&1&0&0 \end{array}\right] \in \mathcal{T}$$

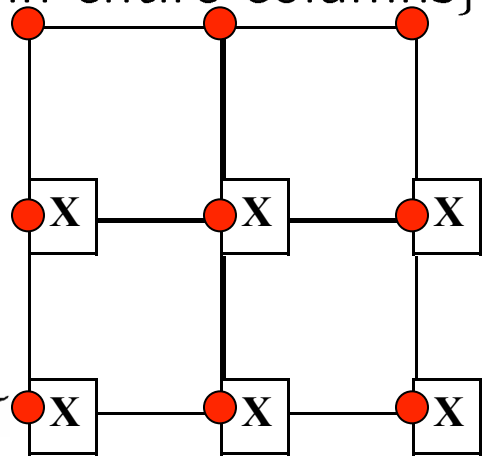


$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # rows with 1′s in entire row

B has an even # columns with 1′s in entire columns$\}$

$\mathcal{S} \subset \mathcal{T}$

$$\left[\begin{array}{cccc|cccc} 1&1&1&1&1&1&0&0 \\ 1&1&1&1&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \\ 0&0&0&0&1&1&0&0 \end{array}\right] \in \mathcal{S}$$



All elements of $\mathcal{S}$ commute with all elements of $\mathcal{T}$

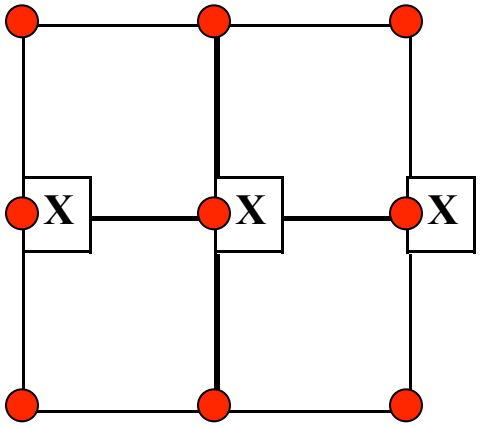
X Columns, Z Rows

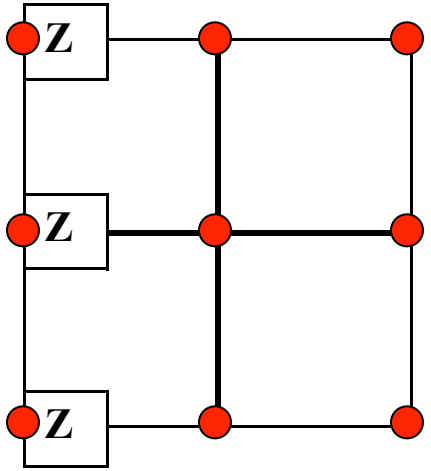
$\mathcal{L} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \mid A$ has an odd # rows with $1'$s in entire columns

B has an odd # colums with $1'$s in entire rows$\}$

$$c\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{L} \qquad \mathcal{L} \notin \mathcal{T}$$

Elements of $\mathcal{L}$ commute with all elements of $\mathcal{T}$





A Little Representation Theory

$\mathcal{S} = \{ \left[\, A \, | \, B \, \right] \;\; | \;\; A$ has an even # rows with 1's in entire row

B has an even # columns with 1's in entire column$\}$

$\mathcal{S}$ commutes with all of $\mathcal{T}$ and $\mathcal{L}$

Label subspaces by $S = \{ \pm 1$ eigenvalues of elements of $\mathcal{S} \}$

$$\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L$$

$\dim = 2^{(n-1)^2}$ $\qquad$ $\dim = 2$

Elements $T \in \mathcal{T}$ act as

$$T = \bigoplus_{s \in S} T_s \otimes I$$

Elements $L \in \mathcal{L}$ act as

$$L = \bigoplus_{s \in S} I \otimes L_s$$

Elements $S \in \mathcal{S}$ act as

$$S = \bigoplus_{s \in S} \pm I \otimes I$$

We will encode a single qubit of in $\mathcal{H}_L$

Quantum Error Correcting Subsystems

$$\mathcal{H} = \bigoplus_{s \in S} \mathcal{H}_T \otimes \mathcal{H}_L$$

Initialize state over subspace labeled by all $s = +1$ eigenvalues of $S \in \mathcal{S}$ as

$$\rho_{arb} \otimes |\psi\rangle\langle\psi|, \; |\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

Error correction for error E need only protect subsystem

$\mathcal{S} = \{ \left[\begin{array}{c|c} A & B \end{array} \right] \;|\; A$ has an even # rows with 1′s in entire row

B has an even # columns with 1′s in entire columns$\}$

$\mathcal{S}_X = \{ \left[\begin{array}{c|c} A & B \end{array} \right] \in \mathcal{S} |$ adjacent rows of A are 1′s$\}$

$\mathcal{S}_Z = \{ \left[\begin{array}{c|c} A & B \end{array} \right] \in \mathcal{S} |$ adjacent columns of B are 1′s$\}$

$$\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \in \mathcal{S}_X \qquad \left[\begin{array}{cccc|cccc} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \in \mathcal{S}_Z$$

Diagnose and Pseudo-Fix

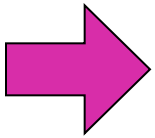
1 Initially in $+1$ eigenstate of elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

2 Error occurs

3 Measure elements of $\mathcal{S}_X$ and $\mathcal{S}_Z$

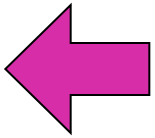
4 Eigenvalues of measurements allow diagnosis and fix modulo $\mathcal{T}$

X error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0 \end{array}\right]$$



Flips to -1 eigenvalues of
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0\\ 0&0&0&0&0&1&1&0 \end{array}\right]$$

$$\left[\begin{array}{cccc|cccc} 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0\\ 0&0&0&0&1&1&0&0 \end{array}\right]$$



Apply fixing procedure
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right]$$

Cumulative error:
$$\left[\begin{array}{cccc|cccc} 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0\\ 0&1&0&0&0&0&0&0 \end{array}\right] \in \mathcal{T}$$

$$T = \bigoplus_{s \in S} T_s \otimes I$$

$$\rho_{err} \otimes |\psi\rangle\langle\psi|$$

Detailing The Fix

$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right]$ | A has an even # of 1′s in columns

B has an even # of 1′s in rows$\}$

violate $\mathcal{T}$ condition

$$\begin{array}{cccc} e & e & o & e \end{array}$$

$$\left[\begin{array}{cccc|cccc} 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} o \\ e \\ e \\ e \end{array}$$

$\leftarrow$ violate $\mathcal{T}$ condition

syndrome:

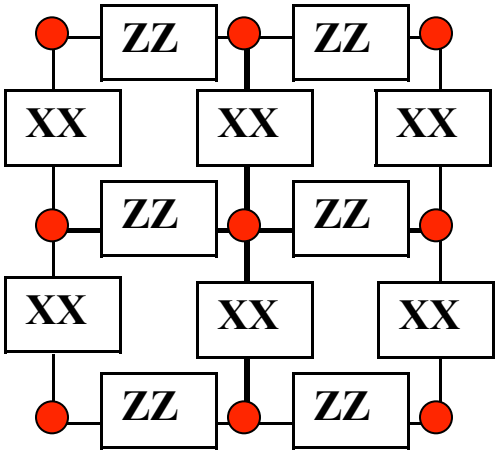
$\mathcal{S}_X = \{\left[\begin{array}{c|c} A & B \end{array}\right] \in \mathcal{S}|$ adjacent rows of A are 1′s$\}$

fix by applying Z's consistent with fewest o's

fixes $\lfloor \frac{n}{2} \rfloor$-1 Z errors

Similar for $\mathcal{S}_Z$

Relation To Physics



Eigenstates of

$$H = \sum_{ij} X_{i,j} X_{i+1,j} + Z_{i,j} Z_{i,j+1}$$

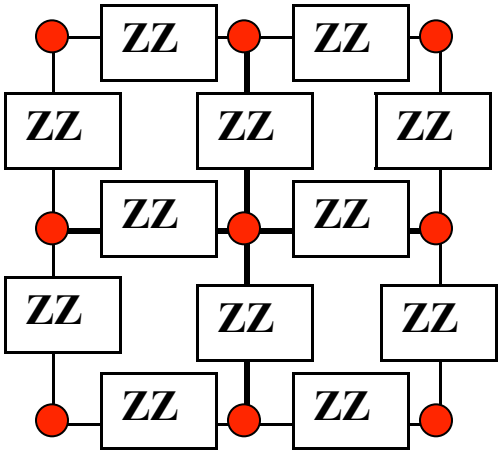
are a quantum error correcting subsystem

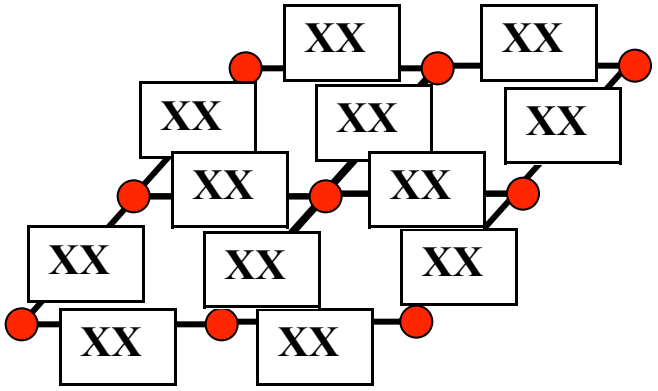
$$H = \bigoplus_{s \in S} H_s \otimes I_2$$

Frinkahedron

n by n by cube of qubits







$$\mathcal{T} = \{c\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an even \# of } 1\text{'s in YZ plane}$$
$$B \text{ has an even \# of } 1\text{'s in XY plane}\}$$

$$\mathcal{S} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an even \# XY planes with } 1\text{'s in entire plane}$$
$$B \text{ has an even \# YZ planes with } 1\text{'s in entire plane}\}$$

$$\mathcal{L} = \{\left[\begin{array}{c|c} A & B \end{array}\right] \mid A \text{ has an odd \# XY planes with } 1\text{'s in entire plane}$$
$$B \text{ has an odd \# YZ planes with } 1\text{'s in entire plane}\}$$

Expectations

$$H = \sum_{ijk} X_{i,j,k}X_{i+1,j,k} + X_{i,j,k}X_{i,j+1,k} + Z_{i,j,k}Z_{i,j+1,k} + Z_{i,j,k}Z_{i,j,k+1}$$

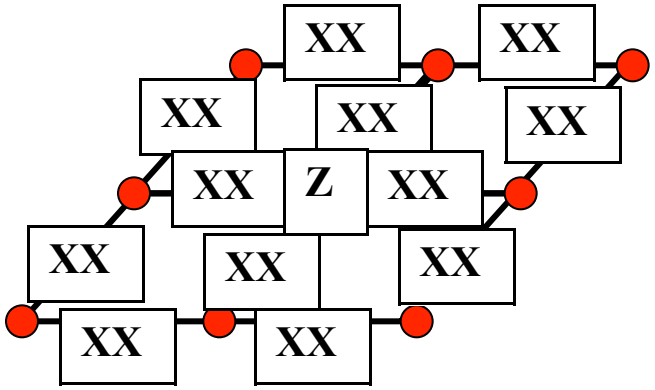
eigenstates are quantum error correcting subsystem

assume phase with two point n.n. correlations

$$\langle X_{i,j,k}X_{i+1,j,k}\rangle = -m_{xx}^{(x)} \qquad \langle Z_{i,j,k}Z_{i,j+1,k}\rangle = -m_{zz}^{(y)}$$

$$\langle X_{i,j,k}X_{i,j+1,k}\rangle = -m_{xx}^{(y)} \qquad \langle Z_{i,j,k}Z_{i,j,k+1}\rangle = -m_{zz}^{(z)}$$

$$\langle H\rangle = -n^2(n-1)(m_{xx}^{(x)} + m_{xx}^{(y)} + m_{zz}^{(y)} + m_{zz}^{(z)})$$



errors are stringlike excitations
in this “mean field” approximation

Energy proportional to perimeter
self correcting?

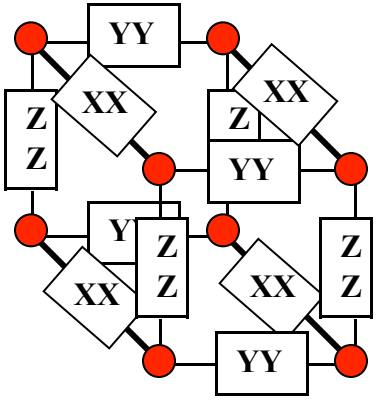
$$\langle H\rangle_{er} = -n^2(n-1)(m_{xx}^{(x)} + m_{xx}^{(y)} + m_{zz}^{(y)} + m_{zz}^{(z)}) + 2m_{xx}^{(x)} + 2m_{xx}^{(y)}$$

Quantum Error Correcting Subsystems

we need not perfectly preserve subspaces

new codes

quantum error correcting subsystems in spin lattice systems



self-correcting quantum memories?

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